Towed-Sensor Surveillance UAV

AMERICAN UNIVERSITY OF THE MIDDLE EAST

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1.0 EXECUTIVE SUMMARY

Objective. Machine Masters will design, build and fly a hand-flown electric aircraft that delivers the sensor in its container (Mission 2), tows it through its light sequence and recovers it (Mission 3), and protects it in the Ground Mission drop, to post the highest total score at Tucson in April 2027.

Approach. The rules became 21 subsystem requirements. A mission model calibrated on the 2022–26 fly-offs flew 162 candidate aircraft through every mission, failed attempts included, against simulated rival fields; completion pays 4.5 of 7.5 points, so reliability carries 55 % of the trade weight. Sensor and container are built for 3–8 kg, with the maximum declared at inspection (+0.42 points over fixing it at design).

Design. A 1.82 m, 1.00 m² high wing with Fowler flaps ($C_{L,\max}$ 1.93), conventional tail, fixed tricycle gear and one 8S tractor stalls at 11.9 m/s at its 14.6 kg Mission 2 mass. It waives Mission 1, carries one 7.4 kg sensor container, tows 3.4 kg and drops the declared sensor from 1.524 m: 6.76 of 7.5 with every mission completed and the field's highest mission total in 31 % of simulated contests, failed attempts counted (first iteration 13 %); a slotted 1.2 m² wing is the fallback.

Execution. Six sub-teams build two airframes of laser-cut ply and balsa on carbon tubes, retire the top risks in 14 bench and flight tests by 15 January, fly the prototype at least 20 times and freeze the design on 19 February; the $38.0k budget, travel included, is fully funded by the university.

DESIGN AT A GLANCE / VARIANT A

WINGSPAN	WING AREA	M2 GROSS MASS	STALL, M2	SCORE, IF DONE	P(TOP OF FIELD)
1.82 m	**1.00** m²	**14.6** kg	**11.9** m/s	**6.76** / 7.5	**0.313**

2.0 MANAGEMENT SUMMARY

2.1 Team Organization

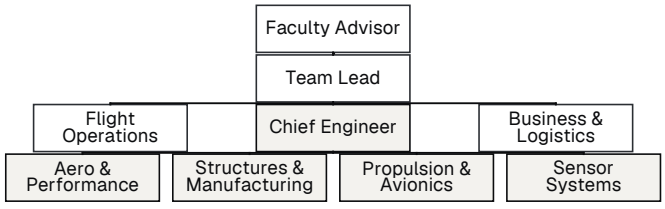


FIG. 01 The chief engineer leads the four technical sub-teams (shaded); each scored subsystem has one owner.

Machine Masters has 21 members in six sub-teams organized by function, mirroring how the aircraft is scored (Fig. 01, Table 01). The faculty advisor signs off spending and flight safety, the team lead owns schedule, budget and rules, and the chief engineer owns the configuration and alone may change a requirement. Sub-teams meet weekly on one version-controlled repository; every decision records its evidence and is reviewed at PDR (28 Nov), CDR with the A/B gate (15 Jan) and freeze (19 Feb). The pilot is an AMA member.

2.2 Schedule and Milestones

The schedule (Fig. 02) runs backward from the fly-off, ordering parts common to both variants first so the A/B decision costs one wing. The prototype is built in December and flies on 6 January; the competition airframe is built from 16 January and flies on 14 February.

2.3 Budget

Table 02 covers the year: the purchase list, a second airframe and wing, 8 kg of tungsten at the $66/kg online average, and for 7 travellers the quoted fares ($1,545

TABLE 01 Sub-teams, what each owns and the skills each requires. Every scored subsystem has exactly one owner.

SUB-TEAM	OWNS	REQUIRED SKILLS
Aero & Performance	Sizing, lap model, stability, CFD	Python modelling; XFoil, AVL, OpenFOAM
Structures & Manufacturing	Wing, Fowler tracks, fuselage, tests	CAD; laser cutting, printing, foam cutting
Propulsion & Avionics	Motor, propeller, pack, fuse, radio	Thrust-stand testing; LiPo safety; ExpressLRS
Sensor Systems	Dart, lights, drum, container, test box	Embedded C; servo drives; cushioning design
Flight Operations	Pilot, observer, crew, checklists	AMA-member pilot; timed staging drills
Business & Logistics	Budget, purchasing, travel, visas	Lead-time planning; cost control; writing

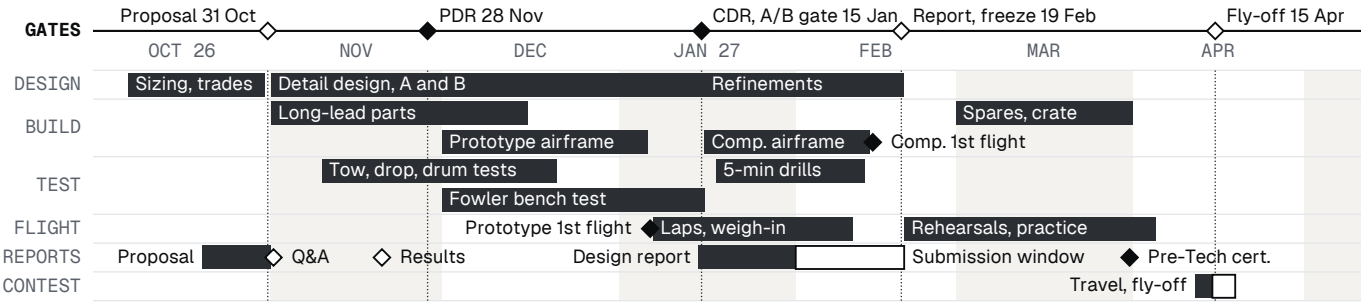


FIG. 02 Decision-gated schedule: long-lead parts first; the prototype flies the Variant A wing in early January; the Fowler test decides A or B on 15 Jan; the configuration freezes with the report on 19 Feb. Outlined: rules dates; filled: team plan.

TABLE 02 Program budget, travel included (USD, rows rounded to $10). Basis: vendor list, quote of 31 Oct 2026, online average, GSA per diem or estimate.

ITEM	BASIS	COST
Prototype airframe, rigs, tools	list	1,760
Comp. airframe, B wing, Fowler test	est.	1,500
Motors, ESCs, radio, servos	list	2,850
8S packs, chargers, 58 V fuses	list	1,950
Sensor, drum, containers, tungsten	list, avg.	1,320
Props, adhesives, consumables	list	460
Crate; Tucson paint, glue	list, est.	250
Airfare, Kuwait–Tucson, 7 people	quote	10,820
Lodging, 4 rooms, 6 nights	quote	2,850
Meals, 7 people, 6.5 days	GSA	3,640
Van, fuel, crate baggage	est.	1,200
US visa fees, 7 people	fee	3,050
Fly-off fee; AIAA, AMA dues	rules, fee	1,420
Contingency, 15 %		4,960
Total		**38,030**
Funding: university, full cost	confirmed	38,030

each), $80/day federal meals and $435 visa fees.[1] Hardware rows are priced line by line at vendor prices, and the long-lead items (fuse, motor, packs, CNC parts) are ordered in November; Business & Logistics tracks spending against each row monthly. A 15 % contingency covers a crash rebuild and outsourced CNC cutting. The university funds the full cost, confirmed.

[1] Lastminute.com quotes, 12–20 Apr. 2027; GSA FY2027 Tucson per diem; U.S. visa fees; six tungsten listings; read 31 Oct. 2026.

3.0 CONCEPTUAL DESIGN APPROACH

3.1 Mission Requirements Decomposition

Each mission score and rule became a numbered subsystem requirement (SR, Table 03); the SR column of Table 09 shows the test that verifies each.

3.2 Scoring Model and Sensitivity Study

The mission score sums four terms, each a completion credit plus a ratio to the best result in the field, our own included:[2]

$$S_{\mathrm{M}} = M_1 + \left[1 + \frac{(W/t)}{(W/t)_{\max}}\right] + \left[2 + \frac{NW_s}{(NW_s)_{\max}}\right] + \left[0.5 + \frac{W_s h}{(W_s h)_{\max}}\right] \le 7.5 \tag{1}$$

Completion is worth 4.5 points and the bonuses at most 3.0, yet in 2026 only 17 of 83 inspected teams completed every mission;[3] winning totals ran 5.61–6.80 in 2022–26. The declared sensor mass W_s sets the Ground Mission product, every Mission 2 container and the Mission 3 ballast cap.

Each design flies 120 random draws of every uncertain input (mass, $C_{L,\max}$, drag, efficiency, density and a lap-time factor from 24 measured flights[4]) against

[2] AIAA, *2026–27 Design, Build, Fly Rules*, 2 Sep. 2026, Secs. 3.1–3.3, 6.3.

[3] AIAA, "DBF Competition Summaries," 2022–2026 (513 team results).

[4] Predicted and measured times in seven DBF design reports, 2021–26; airframe masses of 14 top-three aircraft, 2022–26.

TABLE 03 Mission requirements decomposed into subsystem requirements (SR). † Team requirement beyond the rules.

MISSION	SCORE	KEY CONSTRAINTS	DERIVED SUBSYSTEM REQUIREMENTS
M1 Staging	$M_1 = 1.0$	3 laps in 5 min; may be waived and credited by a successful M2	SR-01 static margin 16 % MAC in every loading†; SR-02 all checks inside the 5-min staging
M2 Delivery	$M_2 = 1 + \frac{W/t}{(W/t)_{\max}}$	Container plus simulators at the declared maximum; TOP up; 5 laps	SR-03 bay for a 288 × 303 × 303 mm container; SR-04 positive restraint; SR-05 CG inside marked limits in every loading
M3 Sensor	$M_3 = 2 + \frac{NW_s}{(NW_s)_{\max}}$	Internal until deployed, before the first downwind turn; OFF, SOLID, FLASHING, OFF; stow to land	SR-06 internal drum, tow ≥ 2.73 m, latched stow; SR-07 three lights, modes over the line, visible in sun; SR-08 no divergent swing at n = 3.24; SR-09 own COTS battery; SR-10 line clear of propeller, tail†; SR-11 same door state every mission
Ground	$GM = 0.5 + \frac{W_s h}{(W_s h)_{\max}}$	6 drops on concrete, $h \leq$ 1.524 m (60 in); ring to bottom < 381 mm (15 in); 5-min rigging	SR-12 container absorbs $W_s h$, sensor designed for 150 g; SR-13 tool-less O/E rigging, flush fittings; SR-14 test box drives all three modes
System	–	Span ≤ 1.829 m (6 ft); TOGW < 24.95 kg (55 lb); ≤ 100 Wh; no autopilot, GPS or camera; arming, fuse, radio rules	SR-15 pilot-in-the-loop only; SR-16 positive latches, no magnets; SR-17 §4.2.5 fail-safe, range check; SR-18 structure for n = 3.24 and tip lift; SR-19 top arming plug ≥ 152 mm aft of the propeller plane, external Rx switch, separate Rx pack, BEC off, fuse ≤ pack rating; SR-20 M2 at full payload within 78.1 Wh usable, stall ≤ 12 m/s; SR-21 ESC, pack within rated temperature†

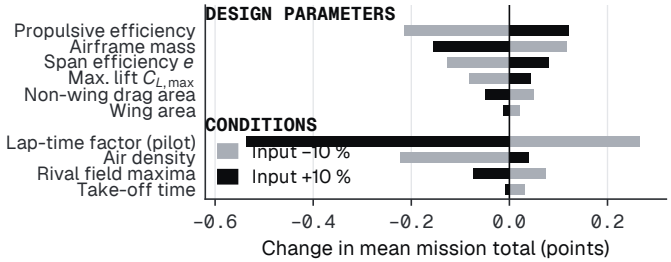


FIG. 03 Mean mission total change for ±10 % on each input: lap time and density, which the team does not set, outweigh every design parameter.

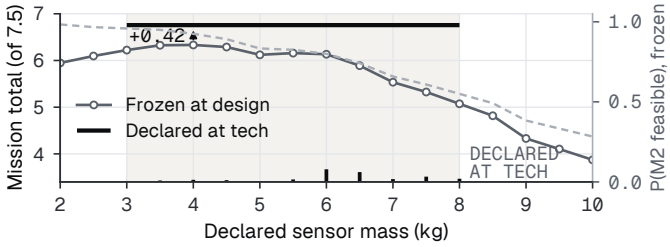


FIG. 04 Declaring at inspection is worth +0.42 over the best mass fixed at design; fixed at 8 kg, Mission 2 is feasible on only 0.55 of airframes. Shaded: 3–8 kg window; bars: masses the draws declare.

5,000 simulated rival fields with attempt success from 2023–26; these inputs describe pilots and staging, not the payload, so they carry over and are refitted to the prototype's laps (TC-10). Efficiency, airframe mass and span efficiency lead the design parameters (Fig. 03; 0.22, 0.16 and 0.13 points per 10 %), and 10 % slower laps cost 0.54. Mission 2 uses 77.9 of the 78.1 Wh usable, leaving the 20 % reserve untouched. Declaring at inspection is the largest lever (Fig. 04): fixed at design the best is 6.33 (4 kg); declared, 6.76 with every mission completed.

3.3 Configuration Trade Studies

Trade weights come from the sensitivity study (completion 55 %, lap performance 17 %, high lift 10 %, airframe mass 7 %, efficiency 5 %, induced drag 4 %, structure and parasite drag 1 % each). The eleven best of 162 combinations of area, flap, declared range and propulsion were re-flown on all 120 draws (Fig. 05) and had to hold Mission 3 completion within 10 % of the best and a worst-10 % total above the first iteration's. The 1.0 m² Fowler wing leads every finalist by 0.012–0.033 in P(top), the chance of the field's highest mission total, each paired 95 % interval above zero.[5] Table 04 gives the other decisions: SD7037 is within 6 % of the best cruise L/D of 13 sections (XFoil, Re 5×10^5) and keeps its maximum lift tripped (UIUC), while the turn-lift leaders

[5] Flaps: R. C. Platt, NACA Rept. 534, 1935; C. J. Wenzinger and T. A. Harris, NACA Rept. 664, 1939; method validated on NACA TN 663. Airfoils: M. S. Selig et al., UIUC Low-Speed Airfoil Tests.

TABLE 04 Configuration decisions. Points: change in mean mission total; matrix: weighted score of 100, team criteria weighted by the sensitivity study (top weights shown).

DECISION	SELECTED	ALTERNATIVES, EVIDENCE
Planform	**Straight**	Delta −1.69, biplane −0.32
Flap	**Fowler, 1.0**	P(top) 0.012–0.033 lower
Airfoil	**SD7037**	L/D 73/92/96 at C_L 0.4/0.6/1.0; trip-proof
		Of 13: E387 62/85/118, MH 114 35/64/119
Wing	**High, 90**	Low 71, mid 53 (bay 30 %, clearance 24 %)
Tail	**Conv., 83**	T 73, twin boom 68 (line clearance 36 %); V: §4.2.5 fail-safe relies on mixing
Gear	**Trike, 85**	Taildragger 76, bicycle 51 (landing 43 %)
Power	**One 8S**	12S −0.02, twin −0.05 in P(top)
Tips	**Plain**	Winglets −0.01 to −0.05
Sensor	**At tech**	Frozen −0.42 (Fig. 04)



FIG. 05 Finalists on the same 120 draws: the 1.0 m² Fowler wing (A) leads; the slotted 1.2 m² wing (B) is the fallback. The axis starts at 0.265; each gap has its paired 95 % interval above zero.

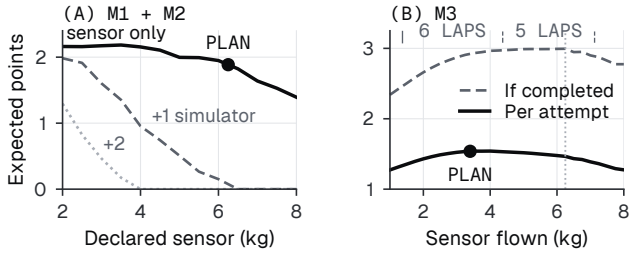


FIG. 06 Payload trades: (a) Mission 1 and 2 points against declared mass, 0 to 2 simulators; (b) Mission 3 points against sensor mass flown; dotted: 6.25 kg declared.

lose 15–52 % at C_L 0.4. Fig. 06 and Table 05 give the payload and drum trades.

3.4 Preliminary Design and Sizing

3.4.1 Sizing method Sizing uses the hot-day density[6] of 1.064 kg/m³ and a lap model of the 152.4 m (500 ft) legs, 180° turns and downwind 360°, with stall

TABLE 05 Deployment-and-recovery trade (scores 1–5). Weights are the mission points at stake; the closed drum wins 99.5–100 % of 5,000 draws with estimates and weights randomised. Boom, wireless, bridle, pulley and spring reel failed the rules or the 20 s limit.

CRITERION	WT.	CLOSED DRUM	OPEN DRUM	FREE SPOOL
Recovery reliability	0.75	5.0	4.4	3.6
Manufacturability	0.15	3.1	5.0	1.7
Weight, volume	0.04	5.0	4.9	2.4
Signal path	0.03	5.0	5.0	5.0
Deploy, stow time	0.03	4.5	5.0	5.0
Weighted total	**1.00**	**4.70**	**4.55**	**3.35**

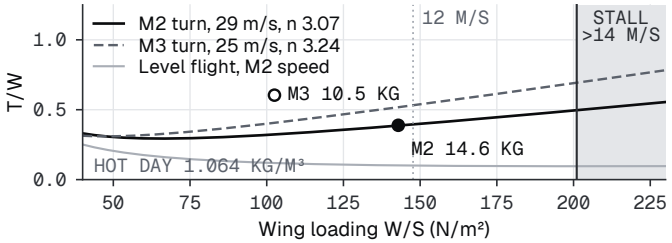


FIG. 07 Mission 2 is sized by the race turn at full thrust; the Fowler flap keeps its 14.6 kg landing under 12 m/s.

≤ 14 m/s, load factor ≤ 4, battery power, the 100 Wh cap and the measured ×1.15 lap factor. Mission 2 sits on its 29 m/s, n = 3.07 turn line at full thrust (Fig. 07); Mission 3 is energy-limited. Of 1,575 motor, propeller and pack combinations, 74 meet the turn limits and 40 were flown through the mission model; the AT7215 with a 20×15E on 8S ranks first, 0.02–0.03 points ahead of a 21×13E or a 12S pack, with 101 N static thrust and 1.4 kW (47 A) mean power in Mission 2.[7]

3.4.2 Aircraft concept Variant A puts a straight wing (Fig. 08, Table 06) on a 0.345 × 0.355 m box fuselage with a conventional tail on a 1.05 m arm. The span sits 9 mm inside the 1.829 m limit (tip caps trimmed at weigh-in) and the area is the 1.00 m² of Fig. 05, so the aspect ratio is 3.31. With the bay centred on the CG and the tow exit 50 mm aft, every loading flies at 16 % static margin; about the CG, AVL gives $C_{m\alpha}$ −0.61, $C_{n\beta}$ 0.076 and $C_{l\beta}$ −0.012 per rad.[8] Variant B changes only the wing (1.2 m², slotted), tail and boom.

[6] 10th percentile at the daily high, 2,878 hourly KTUS observations, 15–18 Apr. 1996–2025, reduced to TIMPA (667 m). Iowa Environmental Mesonet, "ASOS Data Download," retrieved 11 Oct. 2026.

[7] T-Motor and APC published test data; the model is within 3.5 % of a measured 2023 Mission 2 power.

[8] M. Drela and H. Youngren, AVL 3.52; M. Drela, XFoil 6.99; MIT.

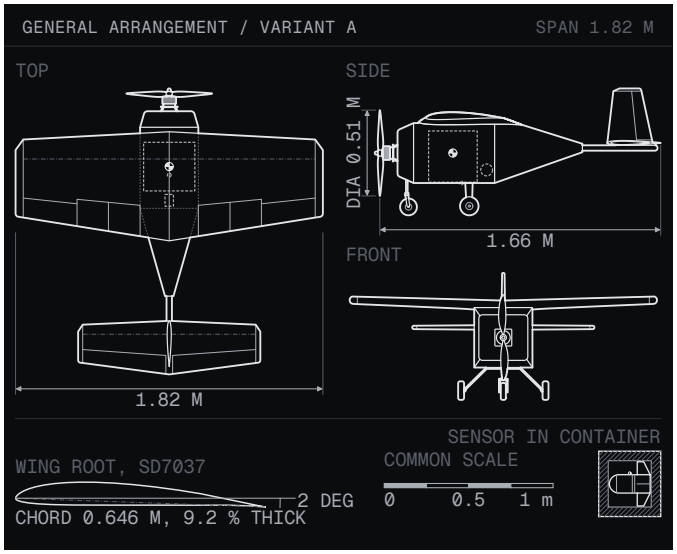


FIG. 08 Variant A to scale (AVL geometry; length from the propeller plane). Dashed: 8 kg container, drum, tow exit; quartered circle: CG. Below: wing root; sensor in its container.

TABLE 06 Design parameters, Variant A. Gross masses and stall speeds are medians of 120 draws, so they do not add up.

PARAMETER	VALUE	UNIT
Span / area / aspect ratio	1.82 / 1.00 / 3.31	m, m²
Mean chord / tail arm	0.555 / 1.05	m
Tail areas, H / V	0.291 / 0.069	m²
Empty mass with pack (fit)	6.96	kg
Gross mass, M2 / M3	14.6 / 10.5	kg
Trimmed landing $C_{L,\max}$	1.93	–
Stall, M2 / M3	11.9 / 10.1	m/s
C_{D0} / L/D, M2 level	0.0216 / 9.8	–
Static margin / $C_{m\alpha}$	16 / −0.61	%, /rad
Static thrust / 8S pack	101 / 97.7	N, Wh
M2 mean power / current	1.4 / 47	kW, A
Tow line, built / minimum	2.83 / 2.73	m

3.4.3 **Sensor and D&R system** The sensor is a finned dart with a tungsten core (190 mm long, 85 mm body, 205 mm fins at 8 kg), towed from an eye above its CG so it flies level;[9] its potted electronics are designed for 150 g, checked by accelerometer in the drop tests, and a 1S pack drives three LED arrays of about 220 cd. A closed-loop servo drum (Table 05) with a slip ring deploys 3.4 kg in about 10 s and stows it in 16.5 s (24.6 s at 8 kg) into a latched dock. In a 3.24 g turn the line carries 107 N at 3.4 kg and 254 N at 8 kg and is rated for four times the 3 g load. Towed 50 mm aft of the CG, the sensor keeps every loading at 16.0–16.1 % static margin; its 1.7 N trailing pull 0.18 m below the CG shifts C_m by under 0.002 (estimate).

TABLE 07 Mission targets: medians of 120 draws (P10–P90); design point in Section 3.5 (M2: 204 s). Scores: means against the expected field, each mission completed.

MISSION	TARGET	SCORE
M1	Waived; credited by the first M2	1.00
M2	7.4 kg container; 231 s (158–287)	1.46
M3	3.4 kg; 20.7 laps×kg (16.6–25.8)	2.85
GM	6.25 kg from 1.524 m: 9.53 kg·m	1.44
Total if completed, mean (P10)		**6.76 (6.25)**
Expected, failed attempts counted		**4.83**
P(highest mission total)		**0.313**

3.4.4 **Container and rigging** Built once for the top of the range, the 288 × 303 × 303 mm container wraps the 205 mm fin span in 41 mm of polyethylene foam per face, keeping the sensor under 150 g in a 1.524 m drop with the ring under 381 mm from every face;[10] pads are re-cut at inspection and simulators trim to 28 g.

3.5 Mission Design Approach and Scoring Targets

Every plan protects completion first (Table 07); speeds and times are the design point, not the medians. **M1:** waived and credited by Mission 2, which lifts a typical team's chance of finishing Mission 3 from 0.40 to 0.72 (65 % per attempt, three attempts); Variant A's own chances are 0.82 (M2) and 0.54 (M3). **M2:** one container and no simulators, since one at 6 kg is feasible in only 0.06 of draws and costs more completion than its W/t earns (Fig. 06a); 29 m/s with 71° turns gives 38.9 s laps, 204 s with take-off (median 231 s). **M3:** lights OFF upwind, SOLID after lap 1, FLASHING after lap 2, OFF and stow after the last lap. Ballast is chosen per attempt (Fig. 06b): the optimum is flat at 3.25–4.25 kg because lighter darts stall slower and climb harder, so 3.4 kg flies 6 laps of 39.9 s at 25 m/s. Charging the stow to the window costs only 0.009 points. **GM:** 6.25 kg from 1.524 m gives 9.53 kg·m, above the expected field best (9.14) in most draws; rigging drilled under 240 s. **Switch rule:** if the Fowler section gives trimmed $C_{L,\max}$ below 1.75, or

[9] J. Barrowman, "Theoretical Prediction of the Center of Pressure," NARAM-8, 1966.

[10] U.S. Dept. of Defense, *MIL-HDBK-304B, Package Cushioning Design*, 1978.

jams more than once in 100 cycles, by 15 January, Variant B is built; it trails A by 0.018 in P(top) when the flap works and leads by 0.044 when it does not.

4.0 MANUFACTURING PLAN

4.1 Preliminary Manufacturing Flow

Airframe, sensor and Ground Mission lanes run in parallel and meet at integration (Fig. 09). The prototype is built in five weeks in December and the competition airframe in four from 16 January, from the same files and jigs. Ribs are laser-cut from 3 mm ply (balsa where unloaded) and bonded onto pultruded carbon-tube spars in a jig; the leading edge is balsa-sheeted and the wing film-covered. The wing splits at the root for the crate and rejoins on a carbon-tube joiner with two bolts, proven by the tip lift (TC-06). Built-up structure beat moulded composites, whose moulds and cure time the schedule cannot carry, and repairs at the field; every part is weighed against its budget.

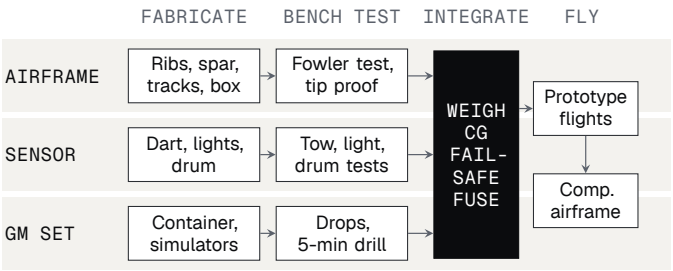


FIG. 09 Airframe, sensor and Ground Mission lanes are each proven on the bench before they meet at integration.

TABLE 08 Components and critical processes: material, method and proof.

COMPONENT	MATERIAL, METHOD	PROOF
Wing	Laser-cut ply ribs, carbon tube	Tip lift
Fowler flaps	CNC Al rail, printed carriage	100 cycles
Fuselage, tail	Ply, carbon boom, balsa	Weigh-in
Gear	Aluminium, faired nose leg	Landings
Drum	Printed drum, servo drive	100 cycles
Dart	Potted tungsten core	Tow, drops
Container	CNC-cut PE foam, ply box	Six drops
Harness	10 AWG, 58 V blade fuse	Fuse duty

4.2 Critical Processes and Technologies

The Fowler track is the one technology the team has never built, so a jigged test section is made first and chooses the variant (Table 08). Its 6061 aluminium rails and the container foam (±1 mm) are CNC-cut by an outside shop, ordered in November with the long-lead parts; carriages and drum are printed and the tungsten core potted in epoxy in-house. The fuse is an 80 A blade rated for 58 V, as the common type is rated for 32 V, below a charged 8S pack's 33.6 V.[11]

5.0 TEST PLANNING

Tests are ordered by the risk they retire and verify the SR column (Table 09); the Fowler test (TC-09, 15 Jan) measures trimmed $C_{L,\max}$ in prototype stalls and cycles the tracks 100 times. Flights build up from taxi and a light, trimmed first flight through stalls and flaps to payload in steps, and stop if control or CG margins, temperatures or the pack reserve are off. The prototype logs at least 20 flights before freeze, each on the checklist below; the weigh-in also checks the span, and TC-14 runs by 10 April. Instruments: thrust-stand load cell, drop-rig accelerometer, ESC telemetry over ExpressLRS (no GPS), video lap timing, scales, CG rig. Top risks (likelihood, consequence): Fowler underdelivers (M, H; TC-09, else B), dart swing (M, H; TC-01), stow failure (H, M; TC-05), fuse opening (L, H; TC-02).

TABLE 09 Test plan (TC-01 to 08 ground, 09 to 14 flight); a failure triggers the Section 3.0 alternative.

TEST, METHOD	SR	PASS CRITERION	DATE
TC-01 Tow rig	08	No divergent swing	Nov
TC-02 Thrust stand	19, 20	$\geq$ 95 N, fuse holds	Nov
TC-03 Drop rig	12	No damage	Dec
TC-04 Sun test	07, 09, 14	Read at 200 m	Dec
TC-05 Drum bench	06	98 % recovery	Dec
TC-06 Tip lift	18	Tips hold 16.1 kg	Dec
TC-07 Weigh, CG	01, 05	Mass ±15, CG ±3 %	Jan
TC-08 Timed drills	02, 13	Under 240 s	Jan
TC-09 Fowler test	20	$C_{L,\max} \geq 1.75$	Jan
TC-10 Timed laps	20	Model ±15 %	Jan
TC-11 Stepped M2	03, 04, 21	5 laps < 300 s	Feb
TC-12 Fail-safe	15–19	§4.2.5, range check	Feb
TC-13 Full M3	06–11	3 modes, stowed	Mar
TC-14 Hot-day run	20, 21	15 % of 97.7 Wh left	Apr

Checklist. Before flight: Rx switch off, plug out; pack, CG and latches checked; §4.2.5 fail-safe and low-power range check; plug in last, at the line. After: plug out first; voltage and temperatures logged; LiPo sacks.

[11] Littelfuse MAXI 32V and MAXI 58V (999 series) datasheets; DigiKey listing 0999080-ZXN, read 29 Oct. 2026.